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software engineering laboratory

An Empirical Evaluation of LLMs for Microservice Bad Smell Detection

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Agenda

- Introduction
- Background
- Study Design
- Expected Results
- Limitations
- Conclusion

Introduction

- Microservices are widely adopted by the industry
- Systems degrades over time due to poor design decisions
- Detection is:
  - Costly
  - Limited
- Opportunity: use LLMs 💡

Background

- Microservices Bad Smells (MBS) are symptoms of poor design
- Examples:
  - No API Gateway
  - Shared Database
  - Cyclic Dependency
  - Shared Libraries

Background

- Previous study:
  - Systematic Literature Review on MBS
  - 104 smells
  - 7 detection tools
  - 15 refactoring strategies for the most discussed ones

Research Questions

- **RQ1:** How effective are LLMs at detecting MBS?
- **RQ2:** How does detection performance vary across different models?
- **RQ3:** How does detection performance vary across different MBS?

Study Design

- Smell selection
  - Focus on specific smells
    - No API Gateway
    - Nano Service
    - Shared Database
    - Microservice Greedy
    - Cyclic Dependency

Study Design

- Models selection
  - Coding-specific models
    - Claude Opus 4.6
    - GPT-5.4
    - Deepseek-coder-v2
    - Qwen-coder-plus
- Dataset
  - Filtered architecture diagrams from an open-source microservices dataset

Study Design

- Prompting strategy
  - Clear definitions
  - Input the diagram as an image
  - Structured output format

```
{
  "smell": "Nano Service",
  "confidence": number,
  "involved_services": string,
  "explanation": string
}
```

Pilot study

- Goal: validate the experimental setup
- Setup:
  - Small subset of architectures
  - Multiple runs per model
- Findings
  - The model performed better separating prompts per smell
  - Very small variability across runs
- Improvements
  - Separate prompts

Execution

- Input: architecture + prompt
- LLM predicts smells
- Output parsed into structured format
- Compare with ground truth
- Multiple runs per configuration
- Metrics:
  - Precision
  - Recall
  - F1-score

Expected Results

- Empirical evaluation of LLMs
- Comparison across models
- Insights on MBS detection
- Contribution:
  - Understanding LLMs in architecture analysis

Conclusion

- Investigating LLMs for MBS and Software Architecture

- Ongoing experiment

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Thank you!

Questions?